

Evidencia — Reconocimiento de la red OT (nmap)

Fase: 3 — Demostración de ataques **Técnica MITRE ATT&CK for ICS:** T0846 — Remote System Discovery
Origen del escaneo: contenedor `kali` (red `red_ot`, 172.21.0.50) **Objetivo:** 172.21.0.0/24
Comando: `docker exec -it kali nmap -sV 172.21.0.0/24`
Fecha: 21 de julio de 2026

Resumen de hosts descubiertos

IP	Host	Puerto	Servicio detectado	Observaciones
172.21.0.1	Gateway de la red	—	—	Todos los puertos filtrados (sin respuesta)
172.21.0.10	openplc	8080/tcp	Werkzeug httpd 0.16.0 (Python 2.7.17)	Interfaz web de gestión de OpenPLC. Puerto 502 (Modbus) aún no visible — pendiente de activar el servidor Modbus en la configuración
172.21.0.11	conpot	2121/tcp, 8800/tcp	Sin fingerprint oficial (nmap no reconoce el servicio)	El honeypot responde imitando un PLC Siemens SIMATIC S7-200 (CP 443-1 EX40) — ver nota abajo
172.21.0.20	wazuh.manager	—	—	Todos los puertos cerrados desde esta red (comportamiento esperado, Wazuh no expone servicios hacia OT)

Nota de interpretación: comportamiento del honeypot

Al recibir una petición HTTP no reconocida, Conpot devuelve una página de error 404 que se identifica explícitamente como un dispositivo industrial real:

```
<TITLE>Not found - Siemens, SIMATIC, S7-200</TITLE>
<h2>CP 443-1 EX40</h2>
```

Esto confirma que el honeypot está emulando correctamente un controlador Siemens ante un escaneo externo, lo cual es el comportamiento esperado de Conpot: parecer un activo ICS legítimo para atraer y registrar intentos de acceso no autorizados. Este registro puede correlacionarse posteriormente como IoA (Indicador de Ataque) en Wazuh.

Salida completa del comando

```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-21 15:55 +0000
Nmap scan report for 172.21.0.1
Host is up (0.000023s latency).
All 1000 scanned ports on 172.21.0.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 3A:34:30:9B:9E:E7 (Unknown)

Nmap scan report for openplc.tfmaquacontrol_red_ot (172.21.0.10)
Host is up (0.000050s latency).
```

Not shown: 999 closed tcp ports (reset)
PORT STATE SERVICE VERSION
8080/tcp open http Werkzeug httpd 0.16.0 (Python 2.7.17)
MAC Address: EA:C5:C3:75:78:01 (Unknown)

Nmap scan report for conpot.tfmaquacontrol_red_ot (172.21.0.11)
Host is up (0.0000050s latency).

Not shown: 998 closed tcp ports (reset)

PORT STATE SERVICE VERSION

2121/tcp open ccproxy-ftp?

8800/tcp open sunwebadmin?

2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at <https://nmap.org/cgi-bin/submit.cgi?new-service> :

=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====

SF-Port2121-TCP:V=7.99%I=7%D=7/21%Time=6A5F967A%P=x86_64-pc-linux-

SF:gnu%r(NULL,17,"200\x20FTP\x20server\x20ready\.\r\n")%r(GenericLines,3A,"200\x20
SF:FTP\x20server\x20ready\.\r\n500\x20Command\x20'\r\n'\x20not\x20understood\r\n")
SF:%r(GetRequest,39,"200\x20FTP\x20server\x20ready\.\r\n500\x20Command\x20'GET'\x20
SF:not\x20understood\r\n")%r(HTTPOptions,3D,"200\x20FTP\x20server\x20ready\.\r\n500
SF:\x20Command\x20'OPTIONS'\x20not\x20understood\r\n")%r(RTSPRequest,3D,"200\x20FTP
SF:\x20server\x20ready\.\r\n500\x20Command\x20'OPTIONS'\x20not\x20understood\r\n")
SF:%r(RPCCheck,32,"200\x20FTP\x20server\x20ready\.\r\n501\x20can't\x20decode\x20com
SF:mand\.\r\n")%r(DNSVersionBindReqTCP,95,"200\x20FTP\x20server\x20ready\.\r\n500\x
SF:20Command\x20'\x00\x1e\x00\x06\x01\x00\x00\x01\x00\x00\x00\x00\x00\x00\x00
SF:\x00\x07VERSION\x04BIND\x00\x00\x10\x00\x03'\x20not\x20understood\r\n")
SF:%r(DNSStatusRequestTCP,6E,"200\x20FTP\x20server\x20ready\.\r\n500\x20Command\x20
SF:'\x00\x0c\x00\x00\x10\x00\x00\x00\x00\x00\x00\x00\x00\x00'\x20not\x
SF:x20understood\r\n")%r(Help,152,"200\x20FTP\x20server\x20ready\.\r\n214-The\x20fo
SF:llowing\x20commands\x20are\x20recognized:\r\n\x20'ABOR'\x20'ALLO'\x20'APPE'\x20'
SF:CDUP'\x20'CWD'\x20'X20'DELE'\x20'HELP'\x20'LIST'\r\n\x20'MDTM'\x20'MKD'\x20'X20'
SF:MODE'\x20'NLST'\x20'NOOP'\x20'PASS'\x20'PASV'\x20'PORT'\r\n\x20'PWD'\x20'X20'QUI
SF:T'\x20'REIN'\x20'REST'\x20'RETR'\x20'RMD'\x20'X20'RNFR'\x20'RNT0'\r\n\x20'SITE'\x
SF:x20'SIZE'\x20'STAT'\x20'STOR'\x20'STOU'\x20'STRU'\x20'SYST'\x20'TYPE'\r\n\x20'US
SF:ER'\r\n214\x20Help\x20command\x20successful.\r\n")%r(SSLSessionReq,32,"200\x20F
SF:TP\x20server\x20ready\.\r\n501\x20can't\x20decode\x20command\.\r\n")%r(Terminal
SF:ServerCookie,32,"200\x20FTP\x20server\x20ready\.\r\n501\x20can't\x20decode\x20co
SF:mmmand\.\r\n")%r(TLSSessionReq,32,"200\x20FTP\x20server\x20ready\.\r\n501\x20can'
SF:t\x20decode\x20command\.\r\n")%r(Kerberos,32,"200\x20FTP\x20server\x20ready\.\r\n
SF:n501\x20can't\x20decode\x20command\.\r\n")%r(SMBProgNeg,32,"200\x20FTP\x20server
SF:\x20ready\.\r\n501\x20can't\x20decode\x20command\.\r\n")%r(X11Probe,63,"200\x20F
SF:TP\x20server\x20ready\.\r\n500\x20Command\x20'L\x00\x0b\x00\x00\x00\x00\x00\x
SF:x00\x00\x00\x00\x00'\x20not\x20understood\r\n");

=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====

SF-Port8800-TCP:V=7.99%I=7%D=7/21%Time=6A5F967A%P=x86_64-pc-linux-

SF:gnu%r(GetRequest,7E,"HTTP/1.1\x20302\x20Found\r\nDate:\x20Tue,\x2021\x20Jul\x20
SF:2026\x2015:49:38\x20GMT\r\nContent-Type:\x20text/html\r\nLocation:\x20/index\.
SF:tml\r\nContent-Length:\x200\r\n")%r(HTTPOptions,9B,"HTTP/1.1\x20200\x20OK\r
SF:\nDate:\x20Tue,\x2021\x20Jul\x202026\x2015:49:38\x20GMT\r\nAllow:\x20GET,HEAD,PO
SF:ST,OPTIONS,TRACE\r\nContent-Length:\x200\r\nConnection:\x20close\r\nContent-Type
SF::\x20text/html\r\n")%r(FourOhFourRequest,13C,"HTTP/1.1\x20404\x20Not\x20Fou
SF:nd\r\nDate:\x20Tue,\x2021\x20Jul\x202026\x2015:49:38\x20GMT\r\nContent-Length:\x
SF:20232\r\n\r\n<HTML>\n\n\x20\x20\x20\x20<HEAD>\n\n\x20\x20\x20\x20\x20\x20\x20\x20
SF:<TITLE>Not\x20found\x20-\x20Siemens,\x20SIMATIC,\x20S7-200</TITLE>\n\n\x20\x20
SF:\x20\x20</HEAD>\n\n\x20\x20\x20\x20\x20\x20<BODY>\n\n\x20\x20\x20\x20\x20\x20\x20<h
SF:2>CP\x20443-1\x20EX40</h2>\n\n\x20\x20\x20\x20\x20\x20\x20\x20<hr>\n\n\x20\x20\x
SF:20\x20\x20\x20\x20\x20\x20
\n\n\x20\x20\x20\x20\x20\x20\x20\x20This\x20resou
SF:rce\x20could\x20not\x20be\x20found\.
\n\n\x20\x20\x20\x20</BODY>\n\n</HTML&
SF:gt;");

MAC Address: BE:A9:FD:AE:D2:FE (Unknown)

Nmap scan report for wazuh.manager.tfmaquacontrol_red_ot (172.21.0.20)

Host is up (0.000020s latency).

All 1000 scanned ports on wazuh.manager.tfmaquacontrol_red_ot (172.21.0.20) are in ignored states.

Not shown: 1000 closed tcp ports (reset)

MAC Address: AE:41:AF:A6:32:A9 (Unknown)